

1. Sort an array in ascending and descending order.

Code:

```
package day_03;

class SortArray {

    public static void main(String[] args) {

        int[] arr = {5, 2, 9, 1, 7};

        int n = arr.length;

        for (int i = 0; i < n - 1; i++) {

            for (int j = 0; j < n - 1 - i; j++) {

                if (arr[j] > arr[j + 1]) {

                    int temp = arr[j];

                    arr[j] = arr[j + 1];

                    arr[j + 1] = temp;

                }

            }

        }

        System.out.print("Sorted array in ascending order: ");

        for (int num : arr) System.out.print(num + " ");

        System.out.println();

        System.out.print("Sorted array in ascending order: ");

        for (int i = n - 1; i >= 0; i--) {

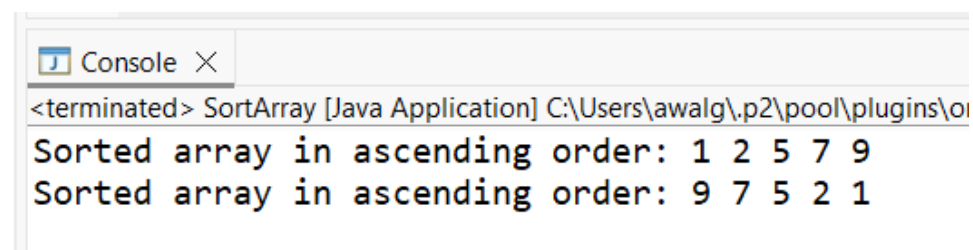
            System.out.print(arr[i] + " ");

        }

    }

}
```

Output:



```
Console X
<terminated> SortArray [Java Application] C:\Users\awalg\.p2\pool\plugins\or
Sorted array in ascending order: 1 2 5 7 9
Sorted array in ascending order: 9 7 5 2 1
```

2. Remove duplicate elements from an array.

Code:

```
package day_03;

class RemoveDuplicates {

    public static void main(String[] args) {

        int[] arr = {1, 2, 2, 3, 4, 4, 5};

        int n = arr.length;

        int[] temp = new int[n];

        int k = 0;

        for (int i = 0; i < n; i++) {

            boolean isDuplicate = false;

            for (int j = 0; j < k; j++) {

                if (arr[i] == temp[j]) {

                    isDuplicate = true;

                    break;

                }

            }

            if (!isDuplicate) {

                temp[k++] = arr[i];

            }

        }

        System.out.print("With duplicates: ");

        for (int i = 0; i < n; i++) {

            System.out.print(arr[i] + " ");

        }

        System.out.println();

        System.out.print("Without duplicates: ");

        for (int i = 0; i < k; i++) {

            System.out.print(temp[i] + " ");

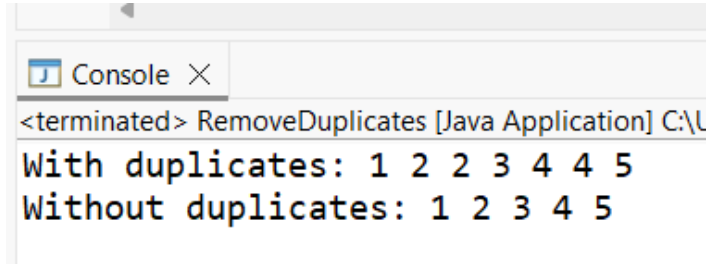
        }

    }

}
```

```
}
```

Output:



```
<terminated> RemoveDuplicates [Java Application] C:\L  
With duplicates: 1 2 2 3 4 4 5  
Without duplicates: 1 2 3 4 5
```

3. Find largest element in each row.

Code:

```
package day_03;  
  
class MaxElementRow {  
    public static void main(String[] args) {  
        int[][] arr = {  
            {1, 5, 3},  
            {9, 2, 8},  
            {4, 6, 7}  
        };  
  
        for (int[] row : arr) {  
            for (int element : row) {  
                System.out.print(element + " ");  
            }  
            System.out.println();  
        }  
  
        for (int i = 0; i < arr.length; i++) {  
            int max = arr[i][0];  
  
            for (int j = 1; j < arr[i].length; j++) {  
                if (arr[i][j] > max) {  
                    max = arr[i][j];  
                }  
            }  
        }  
    }  
}
```

```

    }

    System.out.println("Row " + i + " max: " + max);

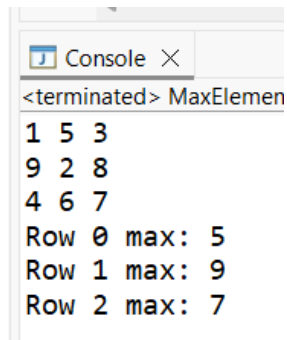
}

}

}

```

Output:



```

<terminated> MaxElemen
1 5 3
9 2 8
4 6 7
Row 0 max: 5
Row 1 max: 9
Row 2 max: 7

```

4. Check if array is palindrome.

Code:

```

package day_03;

class Palindrome {

    public static void main(String[] args) {

        int[] arr = {1, 2, 3, 2, 1};

        int n = arr.length;

        boolean isPalindrome = true;

        for (int i = 0; i < n / 2; i++) {

            if (arr[i] != arr[n - 1 - i]) {

                isPalindrome = false;

                break;

            }

        }

        for (int element : arr) {

            System.out.print(element + " ");

        }

        System.out.println();
    }
}

```

```

    if (isPalindrome)

        System.out.println("The array is Palindrome");

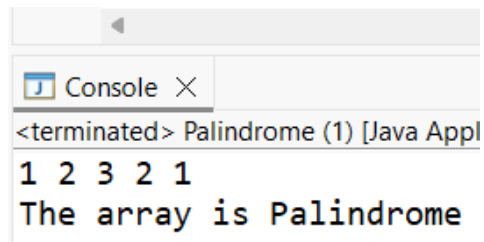
    else

        System.out.println("The array is not Palindrome");

}

}

```



5. Rotate matrix by 90 degrees.

Code:

```

package day_03;

class RotateMatrix {

    public static void main(String[] args) {

        int[][] matrix = {

            {1, 2, 3},

            {4, 5, 6},

            {7, 8, 9}

        };

        System.out.println("Original array: ");

        for (int[] row : matrix) {

            for (int element : row) {

                System.out.print(element + " ");

            }

            System.out.println();

        }

        int n = matrix.length;

        for (int i = 0; i < n; i++) {

            for (int j = i; j < n; j++) {

```

```

        int temp = matrix[i][j];

        matrix[i][j] = matrix[j][i];

        matrix[j][i] = temp;
    }
}

for (int i = 0; i < n; i++) {
    int start = 0, end = n - 1;

    while (start < end) {
        int temp = matrix[i][start];

        matrix[i][start] = matrix[i][end];

        matrix[i][end] = temp;

        start++;

        end--;
    }
}

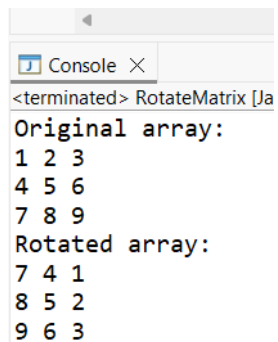
System.out.println("Rotated array: ");

for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        System.out.print(matrix[i][j] + " ");
    }

    System.out.println();
}
}
}

```

Output:



The screenshot shows a console window titled "Console" with a close button. The output text is as follows:

```

<terminated> RotateMatrix [Ja
Original array:
1 2 3
4 5 6
7 8 9
Rotated array:
7 4 1
8 5 2
9 6 3

```